

ENPC IMI Project – Call No. 4

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Agenda

- 1 Method Updates
- 2 Questions and Next Steps

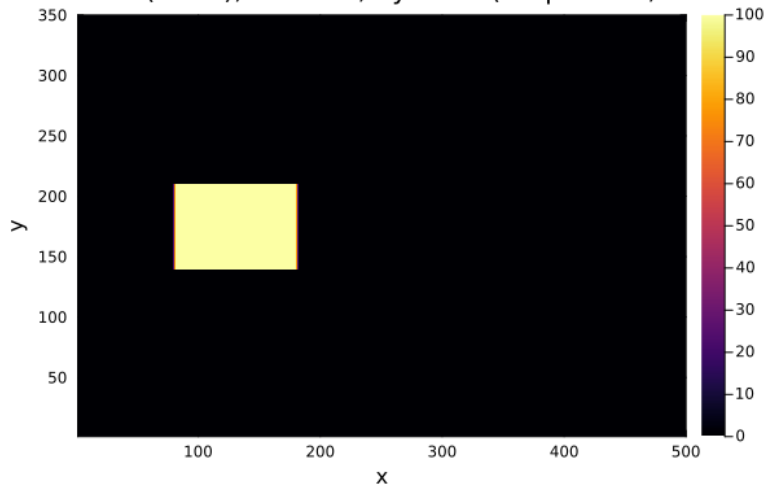
- Verified that non-square grids are supported ($\Delta x \neq \Delta y$).
- Cleaned and unified the experiment scripts and naming.
- Chose script-based validation for now (no Julia Test integration yet).
- Now $0.1 * CFL$ works for all methods and tests

MP5 Fixes and Current Status

- Fixed an indexing bug in the MP5 advection workflow.
- Horizontal and vertical transport tests now behave consistently.
- Rotation cases now run, but error plateaus remain at finer grids.
- Observed some explosion in very irregular velocity fields (SSPRK3 solved it correctly...)

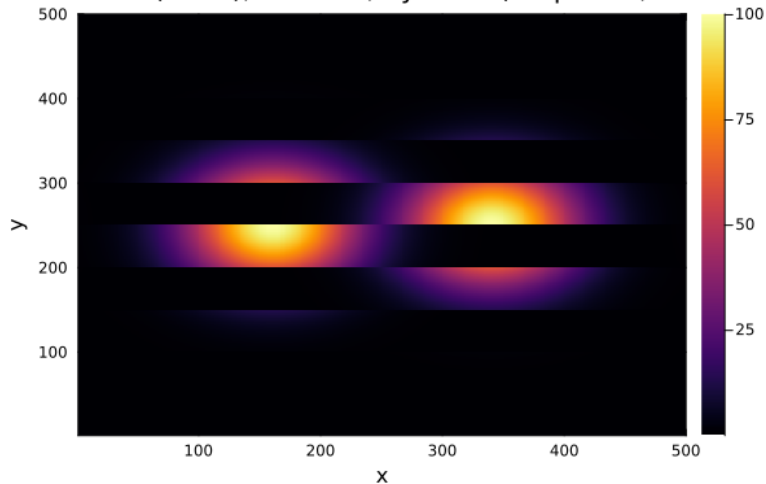
MP5 Fixes and Current Status

ction MP5 (euler), nx=500, ny=350 (Step: 1200, dt=2.5e.



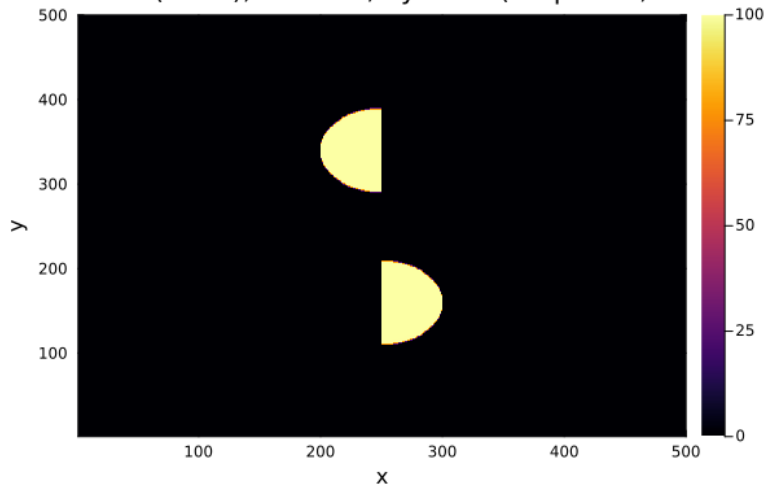
MP5 Fixes and Current Status

action MP5 (euler), nx=500, ny=500 (Step: 900, dt=2.5e-1



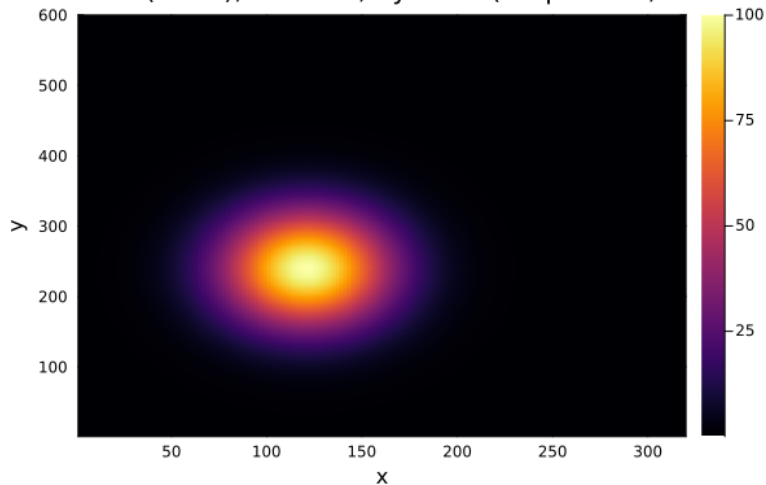
MP5 Fixes and Current Status

action MP5 (euler), nx=500, ny=500 (Step: 900, dt=2.5e-1



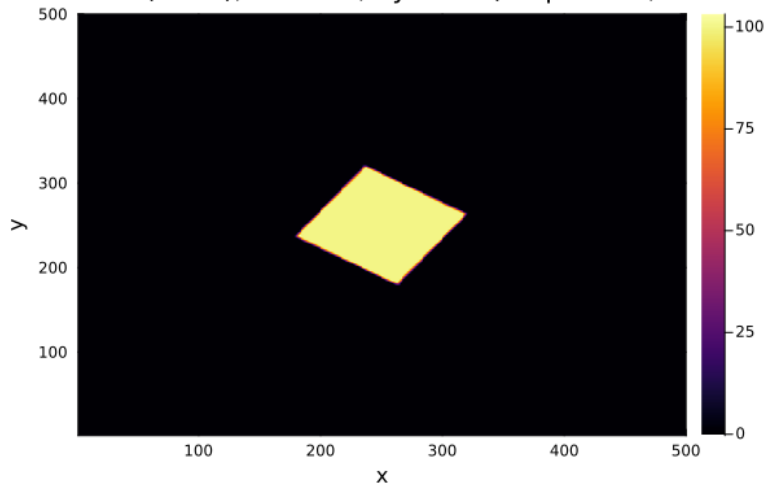
MP5 Fixes and Current Status

Simulation MP5 (euler), nx=320, ny=600 (Step: 1000, dt=2.2e-)



MP5 Fixes and Current Status

Section MP5 (euler), nx=500, ny=500 (Step: 3000, dt=4.0e-)

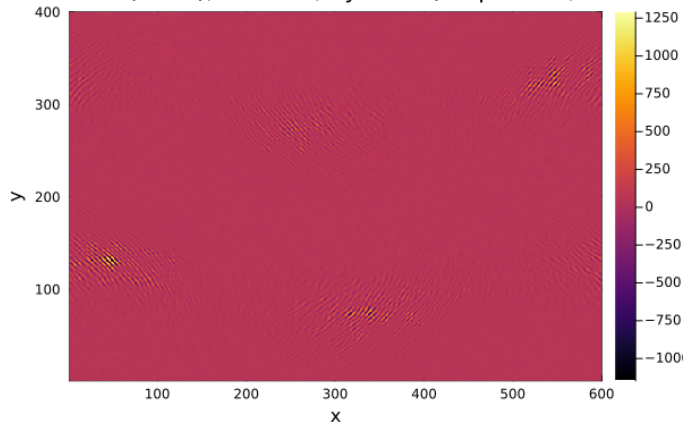


MP5 Fixes and Current Status

Initialized as random noise.

$$v_x(x, y) = 0.5 \sin(2\pi y), \quad v_y(x, y) = 0.5 \cos(2\pi x)$$

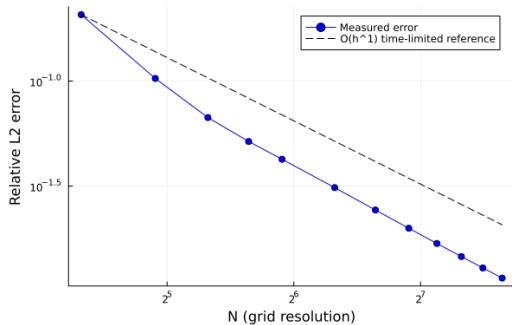
Simulation MP5 (euler), nx=600, ny=400 (Step: 1800, dt=2.0e-5)



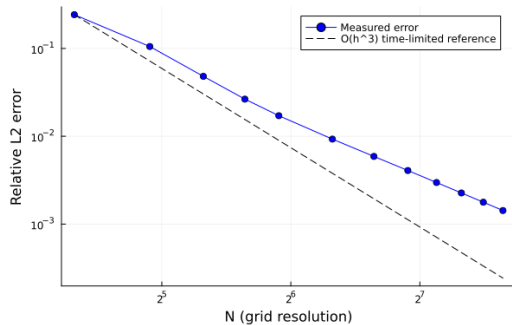
- Implemented $\text{MP5} + \text{SSPRK33}()$ through OrdinaryDiffEq.
- Accuracy improved versus Euler for the same Δt , with higher compute cost per step.

SSPRK3 Integration (SciML)

MP5 Rotating Gaussian (Euler)

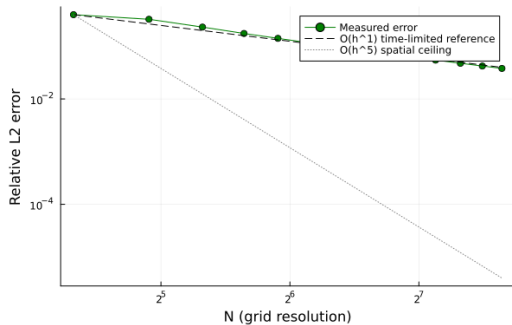


MP5 Rotating Gaussian (SSPRK3)

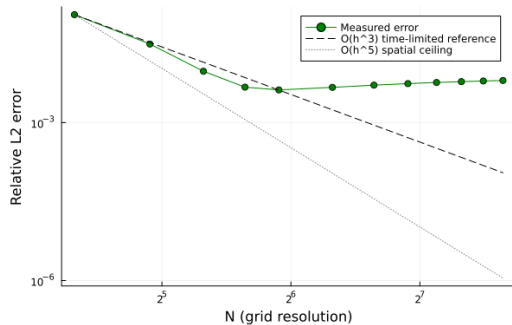


SSPRK3 Integration (SciML)

MP5 Blob Translation (Euler)



MP5 Blob Translation (SSPRK3)

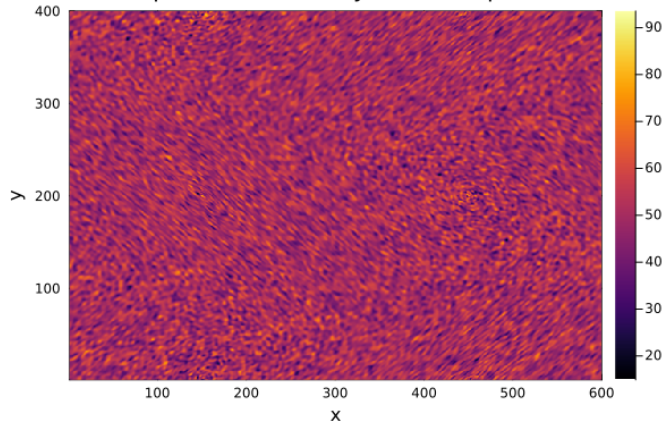


SSPRK3 Integration (SciML)

This got better!

$$v_x(x, y) = 0.5 \sin(2\pi y), \quad v_y(x, y) = 0.5 \cos(2\pi x)$$

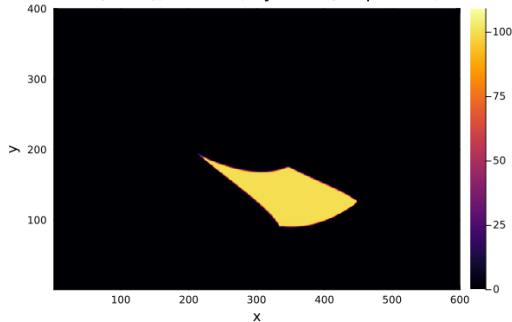
Integration MP5 (ssprk3), nx=600, ny=400 (Step: 1800, dt=2.0e-6)



SSPRK3 Integration (SciML)

On same sinusoidal field

Plot of MP5 (euler), $n_x=600$, $n_y=400$ (Step: 1800, $dt=2.0e-6$)



Plot of MP5 (ssprk3), $n_x=600$, $n_y=400$ (Step: 1800, $dt=2.0e-6$)

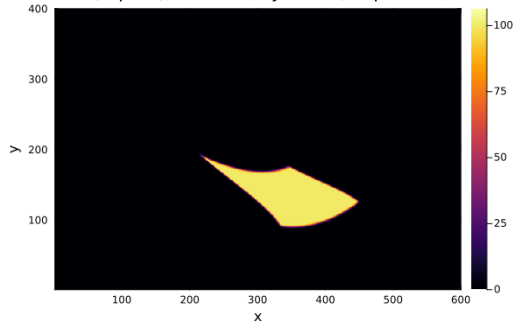


Figure: Both seem to have raising values, but with MP5 (right) slightly better than Euler (left)

Implicit Diffusion with D_{xy} : 9-Point Stencil

- Reworked the tensor diffusion operator to include cross-derivative terms.
- Moved from simplified assumptions ($D_{xx} = D_{yy}$, no cross term) to full (D_{xx}, D_{yy}, D_{xy}) coefficient functions.
- Implemented and tested a 9-point stencil formulation for ∂_{xy} -coupling.

Implicit Diffusion with D_{xy} : 9-Point Stencil

$$\partial_{xy} q = \partial_x(\partial_y q)$$

$$\partial_y q|_{i,j} \approx \frac{q_{i,j+1} - q_{i,j-1}}{2\Delta y}$$

$$\partial_x(\partial_y q)|_{i,j} \approx \frac{\frac{q_{i+1,j+1} - q_{i+1,j-1}}{2\Delta y} - \frac{q_{i-1,j+1} - q_{i-1,j-1}}{2\Delta y}}{2\Delta x}$$

$$= \frac{q_{i+1,j+1} - q_{i+1,j-1} - q_{i-1,j+1} + q_{i-1,j-1}}{4\Delta x \Delta y}$$

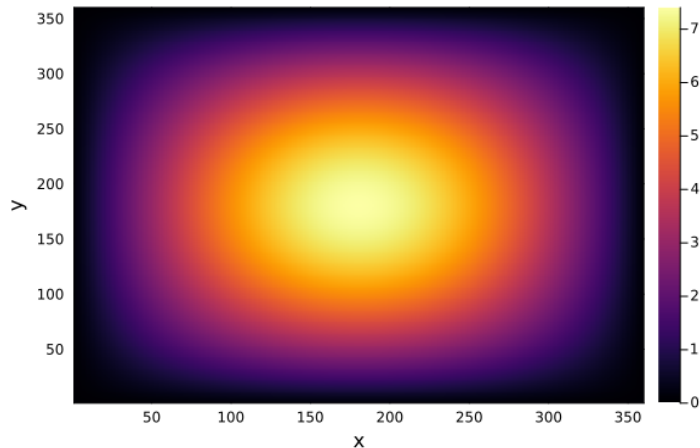
$$\frac{1}{4\Delta x \Delta y} \begin{bmatrix} +1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & +1 \end{bmatrix}$$

Implicit Diffusion with D_{xy} : 9-Point Stencil

Isotropic diffusion

$$D_{xx} = 0.6, \quad D_{yy} = 0.6, \quad D_{xy} = 0$$

r FP diffusion implicit, nx=360, ny=360 (Step: 20, dt=5.0e)

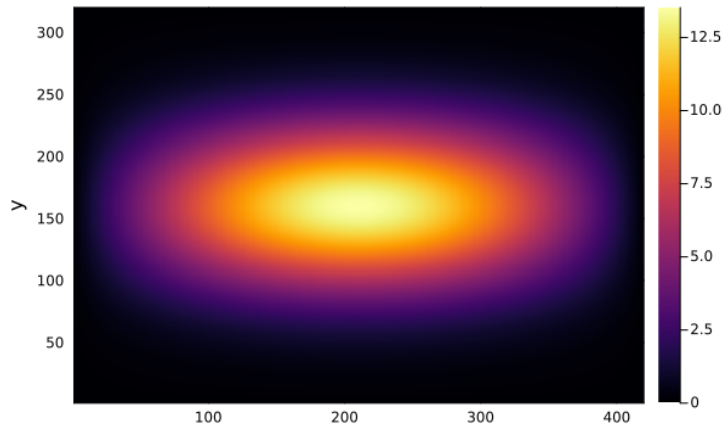


Implicit Diffusion with D_{xy} : 9-Point Stencil

Anisotropic (axis-aligned)

$$D_{xx} = 1.0, \quad D_{yy} = 0.2, \quad D_{xy} = 0$$

1 FP diffusion implicit, nx=420, ny=320 (Step: 25, dt=2.0e-5)

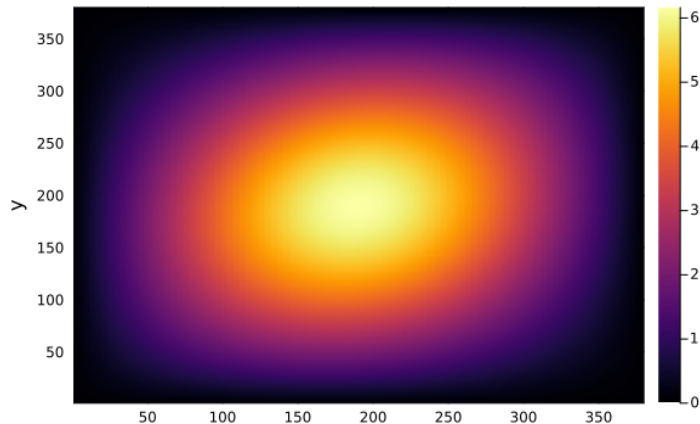


Implicit Diffusion with D_{xy} : 9-Point Stencil

Constant cross diffusion

$$D_{xx} = 0.9, \quad D_{yy} = 0.9, \quad D_{xy} = 0.25$$

1 FP diffusion implicit, nx=380, ny=380 (Step: 30, dt=1.5e-4)

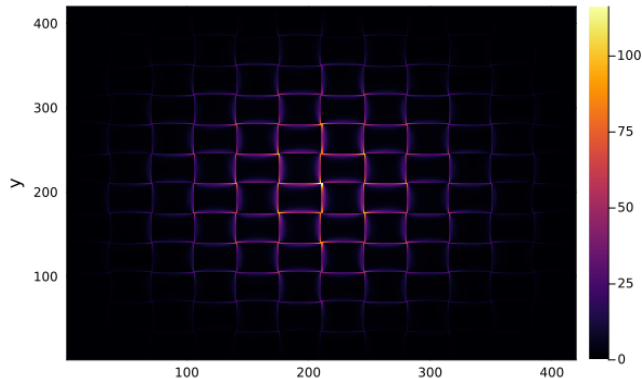


Implicit Diffusion with D_{xy} : 9-Point Stencil

Checkerboard tensor (discontinuous + mixed)

$$D_{xx}, D_{yy}, D_{xy} = \begin{cases} (1.8, 0.08, 0.18) & \text{even cells} \\ (0.08, 1.8, -0.18) & \text{odd cells} \end{cases}$$

r FP diffusion implicit, nx=420, ny=420 (Step: 70, dt=3.0ε)

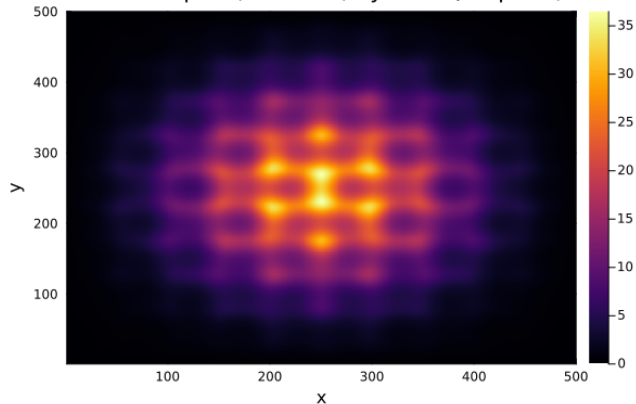


Implicit Diffusion with D_{xy} : 9-Point Stencil

$$D_{xx}(x, y) = 0.7 + 0.25 \sin^2(10\pi x), D_{yy}(x, y) = 0.7 + 0.25 \cos^2(10\pi y),$$

$$D_{xy}(x, y) = 0.25 \sin(12\pi x) \sin(12\pi y)$$

r FP diffusion implicit, nx=500, ny=500 (Step: 80, dt=2.0ε



Strang Splitting (Status and Plan)

- Should I implement the multidimensional strang splitting even if advection works?
- Or only do it for the IMEX at a later stage? $A(\Delta t/2) \rightarrow D(\Delta t) \rightarrow A(\Delta t/2)$.
- Current status: Not done yet

- Is C_μ spatially dependent, i.e., $C_\mu(x, y)$, in the target model?
- Best **validation** strategy for the tensor diffusion solver: find full manufactured solution with nonzero D_{xy} ?
- For Strang splitting, should we first split advection-diffusion only, or also always split 2D advection directionally?

Open Questions

- Is C_μ spatially dependent, i.e., $C_\mu(x, y)$, in the target model?

$$\frac{\partial C}{\partial t} = -\frac{\partial F_x C}{\partial x} - \frac{\partial F_y C}{\partial y} + \frac{\partial^2 D_{xx} C}{\partial x^2} + \frac{\partial^2 D_{yy} C}{\partial y^2} + \frac{\partial^2 D_{xy} C}{\partial x \partial y} \quad (1)$$

$$F_x(x, y) = \sum_{\mu \in \mathcal{M}} m (\beta_\mu C_\mu - \alpha_\mu) \quad (2)$$

$$F_y(x, y) = \sum_{\mu \in \mathcal{M}} q (\beta_\mu C_\mu - \alpha_\mu) \quad (3)$$

$$D_{xx}(x, y) = \frac{1}{2} \sum_{\mu \in \mathcal{M}} m^2 (\beta_\mu C_\mu + \alpha_\mu) \quad (4)$$

$$D_{yy}(x, y) = \frac{1}{2} \sum_{\mu \in \mathcal{M}} q^2 (\beta_\mu C_\mu + \alpha_\mu) \quad (5)$$

$$D_{xy}(x, y) = \sum_{\mu \in \mathcal{M}} m q (\beta_\mu C_\mu + \alpha_\mu) \quad (6)$$

- Is C_μ spatially dependent, i.e., $C_\mu(x, y)$, in the target model?
- Best **validation** strategy for the tensor diffusion solver: find full manufactured solution with nonzero D_{xy} ?
- For Strang splitting, should we first split advection-diffusion only, or also always split 2D advection directionally?

Next steps

Does this make sense?

Did we miss any needed numerical solvers?

Diffusion & Cross-Derivatives (D_{xy})

- Review Kurganov Eq./Sec. 4.20 in detail.
- Test whether that formulation can be adapted/improved for our D_{xy} term.

Physical Model Implementation

- Start with simplified thermodynamics: ignore α (emission) terms initially.
- When adding α , bound $E(m, q)(x)$ between $-m$ and m to avoid exponential blow-up.
- Avoid zero-origin initialization: use small, nonzero concentrations away from $(0, 0)$.
- Implement full Fokker-Planck loop with dummy coefficients first, then calibrate with real parameters.